Chapter 12
The Quick Mild Cognitive Impairment Screen (Q*mci*)

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A.J. Larner (ed.), *Cognitive Screening Instruments*,
DOI 10.1007/978-3-319-44775-9_12

Abstract Differentiating patients with mild cognitive impairment (MCI) from those with subjective memory complaints (SMC) and dementia is important but challenging. Few short cognitive screening instruments with sufficient sensitivity and specificity are available for this purpose in busy clinical settings. The Quick Mild Cognitive Impairment screen (Q*mci*) is a new, short (3–5 min) cognitive screening instrument. Composed of six subtests: orientation, registration, clock drawing, delayed recall, verbal fluency and logical memory, the Q*mci* has excellent accuracy and is highly sensitive and specific at differentiating normal cognition from SMC, MCI and early dementia. The Q*mci* is valid in different settings including memory clinics, general geriatric clinics, movement disorder clinics, and in rehabilitation and general practice. Originally validated in a Canadian sample, it has recently been externally validated in Ireland, Australia, the Netherlands, Turkey and Italy. It is available for clinical and educational use at: http://ageing.oxfordjournals.org/content/early/2012/05/18/ageing.afs059/suppl/DC1 and www.Qmci.org. Cut-off scores, adjusted for age and education, and a new smartphone and tablet computer application (http://www.doctot.com/doctot-apps/dementia-app/) are now available. Further research, with larger sample sizes, is underway to confirm its utility against other short instruments including those designed specifically to detect MCI.

Keywords Screening • Mild Cognitive Impairment • Dementia • Q*mci*

12.1 Introduction

The Quick Mild Cognitive Impairment screen (Q*mci*) is a new, short, clinician-administered cognitive screening instrument (CSI) designed to differentiate mild cognitive impairment (MCI) from subjective cognitive deficits (SCD) and early to mild dementia [1]. Originally designed as a rapid CSI for MCI, it is useful across the cognitive spectrum to screen for cognitive impairment. The Q*mci* has six subtests or subsections: orientation, registration, clock drawing, delayed recall, verbal fluency and logical memory [2]. It is, as its name suggests, quick to score, with a median administration time of under 5 min [2, 3]. The Q*mci* is available as a single 'tear off' sheet with two blank 'clock faces' and a visual scoring aid for its clock drawing subtest on the reverse side. For convenience, validated alternative forms [4] (word groups or versions of the registration and recall task, verbal fluency and logical memory subtests) are included within the same score sheet.

The Q*mci*, developed through an iterative process, is based on another short CSI called the AB Cognitive Screen 135 (ABCS 135) [5]. The ABCS 135, developed by the same research team and published in 2005, is structurally similar to the Q*mci* with five subtests: orientation, registration, clock drawing, delayed recall and verbal fluency, giving it a total score of 135 points [5]. Although the ABCS 135 proved to be a sensitive and brief test to differentiate cognitive impairment from normal cognition, analysis suggested that the weightings of some of its subtests did not enhance the discriminatory properties of the instrument as a whole for MCI [6]. Further, while sensitivity was high, specificity was relatively low. For these reasons the Q*mci* was developed to enhance the sensitivity but particularly the specificity of

the ABCS 135 for MCI. To do this, the weightings of subtests that maximized sensitivity and specificity (delayed recall and verbal fluency) were increased relative to the total score. A new subtest called logical memory, which was scored in parallel to the ABCS 135 during a trial period, was added [1, 4]. The re-weighted instrument, now scored out of 100 points and initially called the ABCS 100, was christened the Q*mci*.

12.2 Q*mci* Screen Scoring and Administration Guidelines

The Q*mci* includes six subtests, covering at least five cognitive domains: orientation, working memory (registration), semantic memory (categorical verbal fluency), visuospatial/executive function (clock drawing) and two tests of episodic memory, (delayed recall and logical memory). Through the re-weighting of its subtests and the addition of logical memory, it places greater emphasis on verbal memory than its predecessor, the ABCS 135. The Q*mci* has a short administration time that if scored according to the guidelines should not take more than 5 min to complete. Each of the Q*mci* subtests including their cognitive domains and administration guidelines are described below and presented in Table 12.1 and Fig. 12.1. Detailed scoring instructions are available on request from the authors or at www.Qmci.org.

Table 12.1 Scoring instructions and timings of the Quick Mild Cognitive Impairment screen (Q*mci*)

Q*mci* subtest	Cognitive domain	Description	Timing	Score
Orientation	Orientation	Five questions; What country, year, month, day, and date?	1 min	10
Registration	Working memory	Five word registration with three alternative word groups	30 s	5
Clock drawing	Visuospatial/construction	Clock drawing within 1 min	1 min	15
Delayed recall	Episodic memory	Five word recall of the five registered words, recalled in any order	30 s	20
Verbal fluency	Semantic memory/language	Naming task: naming from a category with three alternative forms	1 min	20
Logical memory	Episodic memory	A test of immediate verbal recall for a short story	1 min	30
Total score				*/100*

Quick *Mild Cognitive Impairment* screen (*Qmci*)

Name: DOB: Gender: Years in Education: Date: Time:

1. Orientation **To begin ask 5 questions.** One minute. Give **2 points for correct answer,** 1 if attempted but incorrect, 0 if no attempt.	**What country is it?** ___ / 2 **What year is it?** ___ / 2 **What month is it?** ___ / 2 **What is todays date?** ___ / 2 **What day of the week is it?** ___ / 2	**Score** __/ 10
2. Word Registration To begin say… **"I am going to say 5 words. After I have said these 5 words, repeat them back to me. Are you ready?"** 30 seconds. Give **1 point per word** repeated, in any order, no hints.	**Dog** **Rain** **Butter** **Love** **Door** Alternate word groups include… **Cat** **Dark** **Pepper** **Fear** **Bed** **Rat** **Heat** **Bread** **Round** **Chair**	**Score** __/ 5
3. Clock Drawing **"Use the circle provided over page to draw a clock face, set the time to 'ten past eleven'."** One minute approximately. Give 1 mark for each number, 1 for each hand & 1 for the pivot correctly placed or close to their ideal location. Loose 1 mark for each number duplicated or greater than 12, e.g, 15 or 45, i.e. errors.	**Score:** Numbers Correct + _____/ 12 Errors - _____ Hands + _____/ 2 Pivot + _____/ 1 **Total** + _____/ 15	**Score** __/ 15
4. Delayed Recall **"A few minutes ago I named five words. Name as many of those words as you can remember."** 30 seconds. Recall in any order, within 30 seconds, giving **4 points per word,** no hints.	**Dog** **Rain** **Butter** **Love** **Door** Alternate word groups include… **Cat** **Dark** **Pepper** **Fear** **Bed** **Rat** **Heat** **Bread** **Round** **Chair**	**Score** __/ 20
5. Verbal Fluency **"Name as many animals as you can in one minute. Ready? Go."** One minute. Give half a point per animal named; to a maximum of 40. Accept all 'creatures' including birds, fish, insects etc. Do NOT count suffixes twice, e.g. mouse/mice but allow points for similar names calf, cow, bull.	Alternative forms include: ***fruit & veg*** or ***towns & cities.*** **Score 0.5 x number of animals =** List here, in 'shorthand' if required:	**Score** __/ 20

6. Logical Memory

"I am going to read you ONE short story. After I have finished reading it completely, I want you to tell me as much of the story as you can. OK?"

30 seconds.

Give 2 points per highlighted word, recalled exactly, immediately within 30 seconds, in any order, no hints. Two alternatives are provided.

Story 1		Alternative version 1		Alternative version 2	
The **red**	It was a **hot**	The **brown**	It was a **cold**	The **white**	It was a **warm**
fox	**May**	**dog**	**October**	**hen**	**September**
ran across the	**morning.**	**ran across**	**day.**	**walked across**	**afternoon.**
ploughed	**Fragrant**	the **metal**	**Ripe**	the **concrete**	**Dry**
field.	**blossoms**	**bridge.**	**apples**	**road.**	**leaves**
It was **chased** by	were **forming** on	It was **hunting**	were **hanging** on	It was **followed** by	were **blowing** in
a **brown**	the **bushes.**	a **white**	the **trees.**	a **black**	the **wind.**
dog.		**Rabbit.**		**cat.**	

Score __/ 30

***Qmci* Total score** *___/ 100

*adjust score for age & education (see over).

Administered by:_____

Fig. 12.1 The Quick Mild Cognitive Impairment screen (Q*mci*)-scoring sheet, available at http://content.iospress.com/articles/journal-of-alzheimers-disease/jad150881?resultNumber=2&totalResults=9&start=0&q=o%27caoimh&resultsPageSize=10&rows=10 (© O'Caoimh R, Molloy D. W 2011)

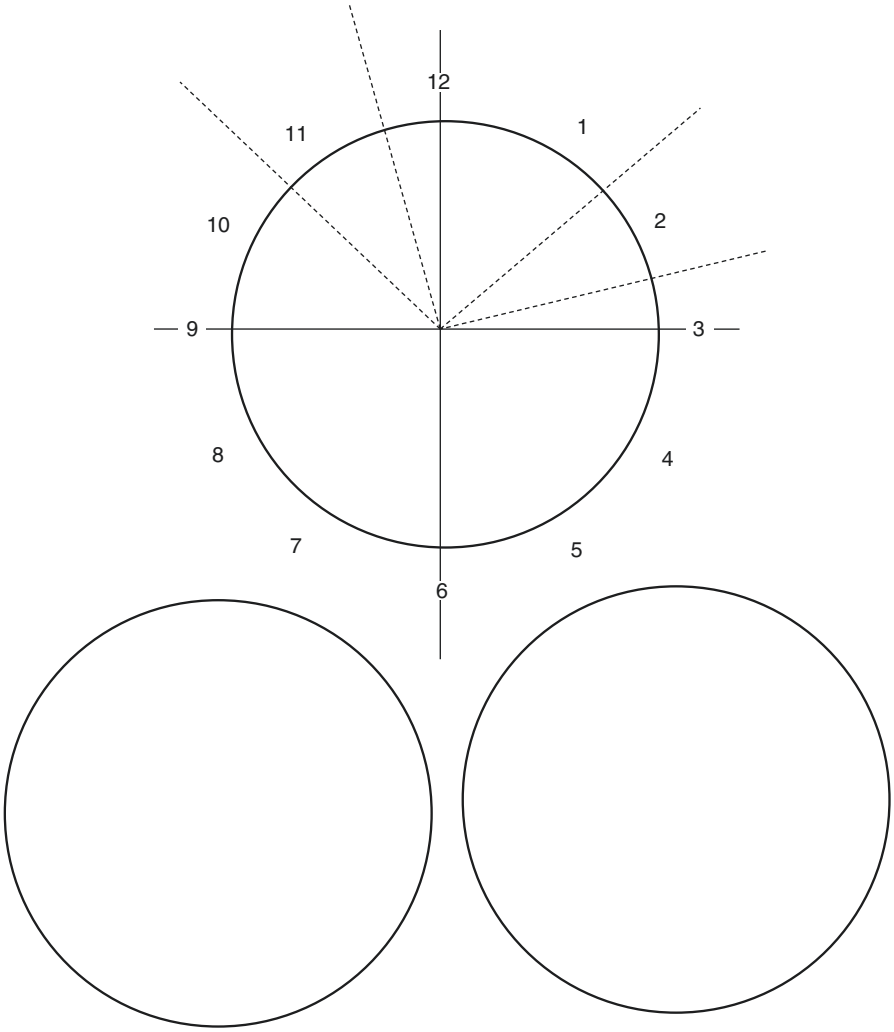

Fig. 12.1 (Continued)

12.2.1 Orientation

The first Q*mci* subtest, orientation, asks five questions and includes tests of orientation in time (What year, month, day, and date?) and place (What country?). It is more heavily weighted towards orientation in time, which is useful in identifying those who warrant more detailed assessment [7] and as a predictor of overall cognitive decline when compared to questions testing orientation to place [8]. Two points are given for the correct answer, one point for wrong answers and zero points for no answer or a conceptually unrelated answer. The timing allows for a maximum of 10

s for each answer to a total time of 30 s. The maximum score is 10 points. Compared with the ABCS 135, the weighting of this subtest was reduced by a factor of 2.5 (from 25 to 10 points) and it now represents just 10 % of the total score i.e. 10 points from a total of 100. Orientation is a poor predictor of MCI with significant ceiling effects [2, 6, 9], and was retained to prevent floor effects so as to allow the instrument to monitor progression in advancing cognitive impairment.

12.2.2 Registration

The second subtest is word registration. It is composed of five items to be repeated back immediately. Three validated alternative word sets are provided [4]. One point is scored for each word recalled after the first reading. If a subject recalls all five, the five items are repeated once before proceeding to the next subtest. If a subject does not repeat all five, the five items are repeated until the subject correctly recalls all items or for a maximum of three trials. The second and third trials do not count towards the score and are there to help the person learn in preparation for the delayed recall subtest. Ten seconds are allowed for recall. The maximum score is five points. Following analysis of the ABCS 135 subtests, registration was reduced by a factor of 5, from, 25 to 5 points.

12.2.3 Clock Drawing

The third Q*mci* subtest is a 1-min clock drawing test (CDT). Clock drawing is a popular short screening test for dementia, in both community [10] and hospital settings [11], and can be scored reliably by both trained and untrained raters [12]. The CDT is a moderately sensitive and specific CSI in its own right (see Chap. 5). The CDT assesses several cognitive domains including visuospatial [13, 14] and executive function [15, 16]. There are several methods of scoring the CDT [15]. The Q*mci* CDT scoring method, based on the technique developed for the ABCS 135, has relatively complex scoring instructions compared to other short CSI that also incorporate the CDT [17, 18]. The Q*mci* CDT scoring instructions are reliable and valid compared to other scoring techniques [19]. Indeed, the increased complexity arguably increases the utility of the subtest [3].

To accommodate the CDT within the Q*mci*, its scoring structure was reduced, by a factor of 2 from 30 points in the ABCS 135 to a new maximum total of 15 points, and the scoring instructions simplified. A blank circle or 'clock face' and transparent scoring template, to be placed over the circle of the completed clock, were provided with the ABCS 135. To simplify scoring for the Q*mci*, new instructions were developed. The subject is still provided with the blank 'clock face', found on the reverse of the two-sided scoring sheet, instructed to '*use the circle provided over page to draw a clock face*' and to set the time to '*ten past eleven*'. One point is given for each number (1–12), for each hand and for the pivot correctly placed at or close

to their ideal location (as denoted on the visual scoring aid accompanying the blank clock face e.g. one point is given for each hand placed between the dashed lines). A single point is lost for each number duplicated or greater than 12, e.g. a 15 or 45, i.e. errors. This provides a total of 15 points. The subject is allowed 1 min.

12.2.4 Delayed Recall

The fourth subtest, five-word delayed recall, tests episodic memory and is also valid as a stand-alone test in dementia [20, 21]. Episodic memory loss occurs early in most dementia subtypes. The Q*mci*'s delayed recall task is based on the five words used in the registration subtest with the CDT functioning as an interval distractor task. The subject is asked to remember the five words, which may be recalled in any order. The Q*mci*'s delayed recall subtest is timed at 30 s with a maximum score of 20 points. Five-word delayed recall adds to the sensitivity of CSIs for MCI, particularly amnestic MCI and is associated with hippocampal atrophy and burden of neurofibrillary tangles in patients with Alzheimer's pathology [22].

12.2.5 Verbal Fluency

The fifth subtest assesses verbal fluency. Verbal fluency facilitates memory retrieval and can be presented as categorical (i.e. semantic, e.g. naming of animals within 1 min) or letter (i.e. phonemic, e.g. naming of words beginning with a designated letter) fluency. Tests of verbal fluency also involve executive control [23]. In the Q*mci*, categorical fluency is assessed with subjects requested to name as many words as possible relating to a named category within 60 s. A half a point is given for each word named to a maximum of 40 words. The final score is rounded up. Words with different suffixes are not counted twice (e.g. fish/fishes, mouse/mice, etc.) but alternate species (e.g. blue jay, robin, sparrow, duck, etc.) are accepted. Alternate validated forms include animals, fruits and vegetables, and cities and towns [4]. The maximum score is 20 points. Compared to the ABCS 135 verbal fluency had its total score reduced from 30 to 20 points, although its overall weighting increased. Patients with Alzheimer's dementia perform less well with categorical fluency than letter fluency, which influenced the decision to include this type of verbal fluency testing within the Q*mci* [24], though both types are abnormal in MCI [25, 26].

12.2.6 Logical Memory

The sixth and final subtest is logical memory, a linguistic test of episodic memory consisting of immediate verbal recall of a short story [27]. Logical memory is a highly sensitive and specific test to differentiate normal cognition from MCI [4] and

is relatively unaffected by age or education [28]. For the Q*mci* version, logical memory is tested using a short story consisting of four sentences which, though not directly connected, provide a coherent 'logical' story. Two points are given for each correct word item recalled verbatim. Only bolded words within each section of the short story need be recalled to score two points. Otherwise the subject scores zero for that word. Each story includes 15 bolded words to provide a maximum score of 30 points. Although no paraphrasing is allowed, recall may be in any order. In total, 30 s are allowed for administration and 30 s for response. Again validated alternatives are available [4].

12.3 Validation of the Q*mci* Screen

The Q*mci*, like the ABCS 135, was originally developed in a Canadian population. The index validation compared the Q*mci* with its predecessor, the ABCS 135, and the Standardized Mini-Mental State Examination (SMMSE) [29, 30] in 965 patients and their caregivers (normal controls) attending four memory clinics in Ontario, Canada [1]. The study showed that the Q*mci* has greater accuracy in differentiating MCI from normal controls than the SMMSE with an area under the receiver operating characteristic (ROC) curve (AUC) of 0.86 versus 0.67 ($p<0.001$) respectively [1]. It also showed that the Q*mci* has greater accuracy than the ABCS 135 (AUC of 0.83, $p=0.05$), while all three instruments accurately separated MCI from dementia including mild dementia when this was separated out from those with moderate to severe stage disease [1]. Tables 12.2 and 12.3 present the characteristics of studies validating the Q*mci* and the psychometric properties demonstrated by the instrument in each study, respectively.

12.3.1 Content Validity

Examination of the subtests of the Q*mci*, using the initial validation data set, showed that all subtests differentiated MCI from normal controls. However, as with the ABCS 135 [6], not all subtests did this in a useful way, with AUC values ranging from 0.56 to 0.80. Logical memory was the most accurate and word registration the least accurate subtest. All subtests distinguished MCI from dementia though orientation was now the most accurate with an AUC of 0.88. The Q*mci* showed excellent test-retest reliability with a correlation coefficient of 0.86 [2]. Median Q*mci* subtest scores, expressed as percentages according to diagnostic classification (normal controls, MCI and dementia), are presented in Fig. 12.2 and show the floor and ceiling effects of the individual Q*mci* subtests.

Table 12.2 Characteristics of participants included in studies validating the Quick Mild Cognitive Impairment screen (Q*mci*)

Country	Language	Setting	Sample size	Sex % Female	Age Median[a] ± IQR	Education Median ±IQR	Reference
Canada	English	Memory Clinic	965	57 %	71 ± 15	13 ± 6	O'Caoimh et al. 2012 [1]
Ireland	English	Movement Disorder Clinic	84	38 %	75 ± 8	12 ± 4	O'Caoimh et al. 2012 [31][b]
Ireland	English	Memory Clinic	551	66 %	76 ± 12	12 ± 4	O'Caoimh et al. 2013 [32], O'Caoimh et al. 2016 [3]
Ireland	English	Geriatric Rehabilitation Unit	82	45 %	81.5 ± 6	12 ± 3	O'Caoimh et al. 2013 [33][b]
Canada	English	Geriatric Clinics (GAT database)	2,113	51 %	77 ± 10	12 ± 5	O'Caoimh et al. 2014 [34][b]
Ireland	English	General Practice	63	67 %	73 ± 17	12 ± 3	O'Caoimh et al. 2015 [35]
Netherlands	Dutch	Geriatric Clinic	90	54 %	72.9 ± 9.1[a]	NA	Bunt et al. 2015 [36]
Australia	English	Geriatric Clinic/Community Clinic	222	52 %	76 ± 13	11 ± 3	Clarnette et al. 2016 [37]
Turkey	Turkish	Geriatric Clinic	100	65 %	75.4 ± 6.9[a]	5 ± 8	Yavuz et al. submitted
Italy	Italian	General Practice	62[c]	45 %	76 ± 9	14 ± 7	Unpublished

IQR Interquartile range, *NA* Not available, *GAT* Geriatric Assessment Tool database
[a]Mean/Standard deviation
[b]Additional data yet unpublished
[c]Ongoing data collection

Table 12.3 Comparison of the psychometric properties of the Quick Mild Cognitive Impairment Screen (Q*mci*) between studies validating the instrument in different countries and settings

Country	Setting	Validated against	Prevalence of cognitive impairment %	Reliability $r = x$		Accuracy (Area under curve)		Reference
				Test-retest	Inter-rater	MCI v Controls ([a]SMC)	MCI v Dementia	
Canada	Memory Clinic	SMMSE, ABCS 135	35 %	0.86	NA	0.86	0.92	O'Caoimh et al. 2012 [1]
Ireland	Movement Disorder Clinic	MoCA	76 %	NA	NA	0.92	0.87	O'Caoimh et al. 2012 [31][b]
Ireland	Memory Clinic	MoCA, 6CIT	79 %	NA	0.97	0.90 (0.81[a])	0.95	O'Caoimh et al. 2013 [32], O'Caoimh et al. 2014 [38][b], O'Caoimh et al. 2016 [3]
Ireland	Geriatric Rehab Unit	MoCA	57 %	NA	0.77	0.76	0.72	O'Caoimh et al. 2013 [33][b]
Canada	Geriatric Clinics	SMMSE	88 %	NA	NA	0.76	0.75	O'Caoimh et al. 2014 [34][b]
Ireland	General Practice	MoCA, GPCOG	51 %	NA	0.89	0.91[a]	0.80	O'Caoimh et al. 2015 [35]
Netherlands	Geriatric Clinic	SMMSE	61 %	NA	NA	0.86	0.73	Bunt et al. 2015 [36]
Australia	Geriatric Clinic/ Community Clinic	MoCA	81.5 %	NA	NA	0.91	0.91	Clarnette et al. 2016 [37]
Turkey	Geriatric Clinic	MoCA	68 %	0.92	0.90	0.80	0.89	Yavuz et al. submitted

Country	Setting	Validated against	Prevalence of cognitive impairment %	Reliability $r = x$		Accuracy (Area under curve)		Reference
				Test-retest	Inter-rater	MCI v Controls ([a]SMC)	MCI v Dementia	
Italy	General Practice	SMMSE, MoCA	56 %[c]	NA	NA	NA	NA	Unpublished

SMMSE Standardized Mini-Mental State Examination, *ABCS 135* AB Cognitive Screen, *MoCA* Montreal Cognitive Assessment, *6CIT* Six-Item Cognitive Impairment Test, *GPCOG* The General Practitioner Assessment of Cognition

[a]Patients with Subjective Memory Complaints (SMC)

[b]Additional data unpublished

[c]Ongoing data collection

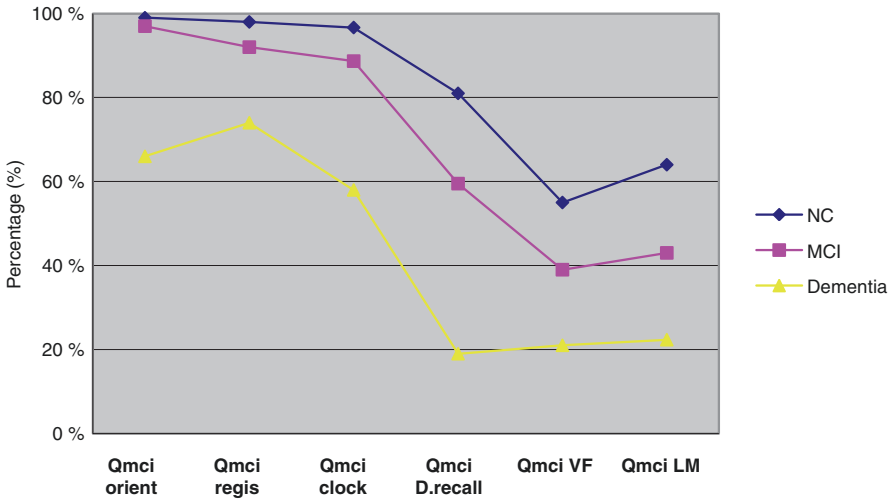

Fig. 12.2 Median Quick Mild Cognitive Impairment screen (Q*mci*) subtest scores expressed as percentages (Image from PhD thesis: https://cora.ucc.ie/handle/10468/2170)

12.3.2 Concurrent Validity

In addition to demonstrating concurrent validity against the SMMSE and ABCS 135 [1], validity has also been demonstrated against other short CSIs including the Montreal Cognitive Assessment (MoCA; Chap. 7) [17], the Six-Item Cognitive Impairment Test (6CIT; Chap. 11) [39] and the General Practitioner Assessment of Cognition (GPCOG; Chap. 10) [40]. External validation of the Q*mci* in Ireland showed that the Q*mci* had higher accuracy (AUC of 0.90 versus 0.80), and comparable sensitivity but greater specificity than the MoCA at their established cut-off scores for differentiating MCI from normal controls [3, 32, 41]. Although the study was underpowered to show superiority, the study reaffirmed the Q*mci*'s shorter administration time, suggesting that where time is limited, such as in busy clinics or general practice, it is a reasonable choice. Concurrent validity was also shown in a subsample of patients attending the same clinic in Ireland against the 6CIT showing that the Q*mci* more accurately identified cognitive impairment (either MCI or dementia), albeit its administration time was twice that of the 6CIT. As expected the Q*mci* best differentiated MCI from normal cognition [38].

A further external validation in primary care (i.e. general practice or family doctors' offices) in Ireland demonstrated that general practitioners and other community-based healthcare professionals, after a brief education session, were able to score the instrument with excellent inter-rater reliability demonstrated compared to trained raters in a memory clinic [35]. Concurrent validity was also shown against the GPCOG and the MoCA. The Q*mci* had statistically significantly greater accuracy than the GPCOG in differentiating SMC from MCI, while its brevity and ease

of administration (no requirement for an informant) further suggest that it is useful in primary care. Most recently, the English version of the Q*mci* has been externally validated in two studies in Australia against the SMMSE [42] and the MoCA [37].

12.3.3 Construct Validity

The construct validity of the Q*mci* against global and neuropsychological test batteries has also been shown. Data from the Doxycycline and Rifampicin for Alzheimer's Disease trial (DARAD) trial, a randomized controlled trial assessing the effects of antibiotics on dementia progression [43], was used to assess internal consistency and the responsiveness of the Q*mci* to change over time [44]. This analysis showed that the Q*mci* had high internal validity, was responsive to change over time and correlated with a detailed neuropsychological battery (the Standardized Alzheimer's Disease Assessment Scale-Cognitive section, ADAS-cog), a global assessment of cognition (the Clinical Dementia Rating scale) and an activities of daily living scale (Lawton-Brody scale). These suggest that the Q*mci* could be substituted for a more detailed neuropsychological instrument in clinical trials [44], the first time that a short CSI has been shown to measure change in cognition function over time in clinical trials. This may be useful, particularly where time or funding is limited or regular detailed monitoring is impractical for raters or unacceptable for subjects. The Q*mci* is now being used in several clinical trials including the FP-7 funded PERsonalised ICT Supported Service for Independent Living and Active Ageing (PERSSILAA; see http://www.perssilaa.eu; project number 610359) [45, 46].

12.3.4 Cut-off Scores

Although normative data are increasingly available for CSIs, few short cognitive screens have established cut-off scores specific to patients presenting with memory loss. To address this, age and education adjusted cut-off scores were developed for the Q*mci* [34]. To increase the sample size available and hence the generalizability of the cut-offs produced, data were pooled from three sources: the original Q*mci* validation data set, the DARAD trial database, and a large outpatient electronic record derived from data contained in the Geriatric Assessment Tool (GAT) database. These data provided a large sample of patients and normal controls from a single country (Canada) with which to develop the cut-off scores. Analysis from this dataset suggests that a cut-off score of <62/100 produces the optimal balance between sensitivity (83 %) and specificity (87 %) for the presence of cognitive impairment (MCI or dementia) using Youden's Index [34]. Using the maximal accuracy approach, a similar cut-off of <63/100 was found, which yielded a

comparable sensitivity (85 %) and specificity (85 %) for cognitive impairment. The suitability of the cut-offs produced was confirmed using data from the external validation in Ireland, with a cut-off score of <62/100 producing a sensitivity of 90 % and specificity of 87 % for cognitive impairment [3]. The cut-off for separating MCI from normal cognition increased to <67/100 irrespective of the method used to derive the score. Cut-offs were also adjusted for subjects' age and education. These confirmed the requirement to adjust scores, particularly for those aged over 75 years.

12.4 Clinical Utility of the Q*mci* Screen: Use in Different Settings

The Q*mci* is validated in different clinical settings as described in Sect. 12.3, including memory clinics [1, 3], geriatric outpatients [34, 36, 37], in the community (general practice or community outreach team) [35, 42], a university hospital rehabilitation unit [33] and a movement disorder clinic [31]. In addition to these, the Q*mci* has been used as an outcome measure in several clinical trials [44, 46–48] and a case control study of a 'memory gym' intervention in MCI [49]. Analysis of data from the GAT database in different dementia subtypes and depression shows that the Q*mci* screen had significantly greater accuracy at differentiating vascular and Parkinson's disease dementia compared with the SMMSE [50]. In that geriatric outpatient sample, while higher AUC values were found, there were no significant differences between the Q*mci* screen and SMMSE in identifying Alzheimer's, Lewy Body, or frontotemporal dementia from subjects with normal cognition, although sample sizes were small and it was not possible to separate MCI subtypes [50].

12.5 Translations of the Q*mci* Screen

To date, apart from English, the Q*mci* has been translated into ten languages (Dutch, Turkish, German, Italian, Portuguese, Polish, Greek, Tamil and Chinese including an adaption for Taiwanese) and these have been externally validated in Dutch [36] and Turkish (unpublished data under review). In the Netherlands, the Q*mci*-Dutch was more sensitive and specific than the widely used SMMSE-Dutch in differentiating MCI from dementia and dementia from normal controls [36]. Similarly, the Q*mci*-TR (Turkish) has shown similar accuracy to the MoCA (Turkish version), albeit with a shorter administration time. Validations are at an advanced stage in Italy and Portugal. Dutch and Italian versions are being used in the PERSSILAA project [45, 46].

12.6 The Quick Memory Check

The Q*mci* was shortened and further reweighted to develop a home, caregiver-administered CSI called the Quick Memory Check (QMC). This short instrument, the first validated caregiver-administered cognitive screen, contains three of the Q*mci* subtests: orientation, verbal fluency and logical memory, and is also scored out of 100 points. Initial validation against the Q*mci* and MoCA, completed by trained raters in clinic, suggests that the QMC is acceptable and can identify cognitive impairment (MCI or dementia), potentially improving the efficiency of busy clinics [51].

12.7 Conclusions and Future Research

This chapter explores the development and the results of the initial validation of the Q*mci*, a new, short CSI for differentiating MCI from SMC and dementia. It presents the concurrent validity of the Q*mci* against a selection of widely used and validated instruments. It also confirms its construct validity against global cognitive and functional scales and the gold-standard outcome measure used in clinical trials, the ADAS-cog.

The place for the Q*mci* in clinical practice is likely to be in community practice. However, the optimal extent, type and benefits of cognitive screening remain uncertain [52–55]. Cognitive screening, especially in busy non-specialized outpatient clinics and in general practice, is limited by the psychometric properties of CSIs in patients who present with SCD. Given this, short, easy to administer, accurate CSIs are required. To date, most studies have assessed the accuracy of screens in highly selected samples, usually patients attending memory clinics where the prevalence of cognitive impairment is generally high. Few instruments have been compared in general practice where the prevalence is low [56] and the utility of and need for these instruments is arguably at its greatest. The Q*mci* may fill this gap but as it is a new instrument, it requires further validation. In particular, its concurrent validity should be demonstrated against detailed neuropsychological assessment and new diagnostic algorithms that take neuroimaging, blood and cerebrospinal fluid results into account [57]. Furthermore, to improve reliability the Q*mci* requires standardization of its scoring instructions, a technique that has improved the scoring of the MMSE [29] and ADAS-cog [58].

Although there is some evidence that the Q*mci* is responsive to change over time [44] and useful in measuring conversion from MCI to dementia [49], it remains to be seen if the Q*mci* is useful in measuring and predicting progression from SCD to MCI and dementia. Normative data are also required to place screening scores in context [59]. A computerized application for smart phones and tablets has recently been developed (http://www.doctot.com/doctot-apps/dementia-app/). Comparing the paper-based Q*mci* to the application is ongoing to confirm convergent validity. External validation of the Q*mci* is also ongoing in other countries, settings and subtypes of cognitive impairment.

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